

The University of Texas at Austin
IN-CLASS WORK 4

M339G Predictive Analytics

April 27, 2026

PROBLEMS.

CONCEPTUAL QUESTIONS

Problem 4.1. (10 points) What is the difference between bagging and random forests? Which one is preferable and why?

Solution. Solutions will vary. The salient point of any response which is to earn credit must be that random forests always generate trees with a smaller randomly chosen number of predictors. Thus, correlation between trees is reduced.

FREE-RESPONSE PROBLEMS.

Problem 4.2. (5 points) Let L be a line in $\mathbb{R}^2$ through the point $\vec{p} = (2, -1)$ in the direction $\vec{v} = (3, 1)$. Provide an example of a normal vector of this line, write down the normal equation, and the standard equation of the form $\beta_0 + \beta_1 x + \beta_2 y = 0$ for the line L . Give an example of a point which is on one side of the line and another which is on the other side of the line. Prove **algebraically** that they are, indeed, on opposite sides of the line.

Solution. An easily found normal vector is $\vec{n} = (-1, 3)$ as

$$\vec{n} \cdot \vec{v} = (-1)(3) + 3(1) = -3 + 3 = 0.$$

Since we know that $\vec{p}$ is on the line L , we have the following normal equation

$$(-1, 3) \cdot (x - 2, y + 1) = 0 \quad \Rightarrow \quad -(x - 2) + 3(y + 1) = 0.$$

Finally, we get the equation

$$-x + 3y + 5 = 0.$$

The origin $(0, 0)$ is “below” the line as

$$-(0) + 3(0) + 5 = 5 > 0.$$

The point $(4, -3)$ is “above” the line as

$$-(4) + 3(-3) + 5 = -8 < 0.$$

Since the expressions have opposite signs, the two points lie on opposite sides of the line.

Problem 4.3. (15 points) Find the point on the plane in $\mathbb{R}^3$ given by

$$2x - y + 4z = 3$$

that is closest to the origin.

Solution. In other words, we are looking for the minimum value of the function

$$f(x, y, z) = x^2 + y^2 + z^2$$

subject to the constraint

$$2x - y + 4z = 3.$$

The Lagrangian is

$$L(x, y, z) = x^2 + y^2 + z^2 + \lambda(2x - y + 4z - 3).$$

The partial derivatives are

$$\frac{\partial L}{\partial x} = 2x + 2\lambda \frac{\partial L}{\partial y} = 2y - \lambda \frac{\partial L}{\partial z} = 2z + 4\lambda \frac{\partial L}{\partial \lambda} = 2x - y + 4z - 3$$

Equating the above to zero, we obtain the following system of equations:

$$2x + 2\lambda = 0 \quad 2y - \lambda = 0 \quad 2z + 4\lambda = 0 \quad 2x - y + 4z = 3$$

From the first equation, we get

$$x = -\lambda.$$

From the second equation, we obtain

$$y = \frac{\lambda}{2}.$$

The third equation yields

$$z = -2\lambda.$$

If we substitute these expressions for x , y and z into the constraint equation, we get

$$2(-\lambda) - \frac{\lambda}{2} + 4(-2\lambda) = 3 \Rightarrow -\frac{21\lambda}{2} = 3 \Rightarrow \lambda = -\frac{2}{7}.$$

Now, it's easy to get

$$x = \frac{2}{7}, \quad y = -\frac{1}{7}, \quad z = \frac{4}{7}.$$

This is the point we were looking for.

The function f 's minimal value is

$$\frac{4 + 1 + 16}{7^2} = \frac{21}{49} = \frac{3}{7}.$$

So, the distance is

$$\sqrt{\frac{3}{7}} = 0.6546537.$$

Problem 4.4. (10 points) Consider the following data set with the explanatory random variable X and the categorical response Y :

X	1	2	2	3	3	4	6	10	12	12
Y	G	B	G	G	B	B	M	B	M	M

The data are split according to the partition by $X < 6$ and $X \geq 6$. What is the total (weighted) cross-entropy, Gini index, and classification error.

Solution. The region where $X < 6$ has weight 0.6. So, the region where $X \geq 6$ has the weight 0.4. For the Gini index, we get

$$0.6(2)(0.5)(0.5) + 0.4(2)(0.25)(0.75) = 0.45$$

For the cross-entropy, we get

$$-(0.6(2)(0.5 \ln(0.5) + 0.4(0.25 \ln(0.25) + 0.75 \ln(0.75))) = 0.6408224$$

Classification error: $\frac{3+1}{10} = 0.4$.

MULTIPLE-CHOICE PROBLEMS.

Problem 4.5. (5 points) *Source: MAS-II, Fall 2019.*

You are given the following three statements about tree-based methods for regression and classification:

- I. The main difference between bagging and random forests is the number of predictors considered at each step in building individual trees.
- II. Single decision tree models generally have a higher variance than random forest models.
- III. Random forests provide an improvement over bagging because trees in a random forest are less correlated than those in bagged trees.

Which one(s) if the above statements are **true**?

- a. I only.
- b. II only.
- c. III only.
- d. I, II and III.
- e. The correct answer is not given above.

Solution. (d)

All three statements are true.

Problem 4.6. (5 points) Consider the following data set with the explanatory random variable X and the categorical response Y :

X	1	2	6	8	12	16	17	20	22	
Y	N	N	L	N	L	N	L	L	L	

Determine which of these splits is/are the best using classification error as the criterion.

- I. $R = \{X \leq 7\}$ and $R^c = \{X > 7\}$
 - II. $R = \{X \leq 10\}$ and $R^c = \{X > 10\}$
 - III. $R = \{X \leq 14\}$ and $R^c = \{X > 14\}$
- a. I only.
 - b. II only.
 - c. I and II only.
 - d. I and III only.
 - e. II and III only.

Solution. (b)

For both I and II, there are 3 misclassifications total whereas there are 2 for II.

Problem 4.7. (5 points) Consider the following statements involving classification trees.

- I. The use of Gini index or cross-entropy may result in split nodes with the same predicted class.
- II. Cross-validation can be used to prune the trees.
- III. In a single node with two possible classes, the Gini index always exceeds the cross-entropy.

Which of the statements above is/are **true**?

- a. I only.
- b. II only.
- c. I and II only.
- d. I and III only.
- e. II and III only.

Solution. (c)

Statement III is incorrect by the graph you were supposed to plot in a homework assignment.

Problem 4.8. (5 points) *Source: SRM Sample Questions.*

Consider the following statements:

- I. Principal Component Analysis (PCA) provide low-dimensional linear surfaces that are closest to the observations.
- II. The first principal component is the line in p-dimensional space that is closest to the observations.
- III. PCA finds a low dimension representation of a dataset that contains as much variation as possible.
- IV. PCA serves as a tool for data visualization.

Which one(s) if the above statements is/are **true**?

- a. I, II and III only.
- b. I, II and IV only.
- c. I, III and IV only.
- d. II, III and IV only.
- e. The statements are all correct.

Solution. (e)

All the statements are true.

Problem 4.9. (5 points) *Source: SRM Sample Questions.*

Determine which of the following statements describe the advantages of using a subset selection process, such as backward selection.

- I. Doing so will likely result in a simpler model.
- II. Doing so will likely improve prediction accuracy.
- III. The results are likely to be easier to interpret

Which one(s) if the above statements is/are **true**?

- a. I only.
- b. II only.
- c. III only.
- d. I, II and III.

e. The correct answer is not given by any of the above.

Solution. According to the SOA, the correct answer is (d). I believe only I and III are true since the reduction of variance - if the truly irrelevant variables are removed - speaks to precision and not accuracy.

Problem 4.10. (5 points) *Source: SRM Sample Questions.*

Which one(s) if the following statements is/are **true**?

- I. If the number of predictors used at each split is equal to the total number of available predictors, the result is the same as using bagging.
 - II. When building a specific tree, the same subset of predictor variables is used at each split.
 - III. Random forests are an improvement over bagging because the trees are decorrelated.
- a. None.
 - b. I and II only.
 - c. I and III only.
 - d. II and III only.
 - e. I, II and III.

Solution. (c)

All the statements are true.